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Reproductive Biology and Parental Investment in *Aviofauna militaris*: Evidence of Extended Post-Hatching Care in *Su-57 Felon* Populations

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ABSTRACT

Long-term, non-invasive monitoring of a previously undocumented breeding colony of Su-57 Felon in the boreal forest zone of northern Eurasia has yielded the first conclusive evidence of prolonged parental care and altricial development in any member of the clade Aviofauna militaris. Over a 19-month observation period, adult individuals were documented returning repeatedly to a complex nest structure, provisioning juveniles with both kinetic munitions and processed hydrocarbon-based nutrients. Juveniles exhibited delayed fledging, with first sustained flight recorded only after 10–13 months of nest residence. These findings indicate a K-selected reproductive strategy characterized by high parental investment, contrasting sharply with the r-selected strategy inferred for earlier-diverging taxa such as MiG-21. The data necessitate revision of current population models and have direct implications for the conservation status of fifth-generation stealth platforms.

Keywords: Aviofauna militaris, Su-57 Felon, parental care, altricial development, K-selected reproduction, military ornithology, nesting ecology, stealth platform conservation

1. INTRODUCTION

The reproductive biology of combat aircraft has remained one of the most significant gaps in our understanding of technobiological systems. While substantial progress has been made in elucidating their unique respiratory architecture, ammogenic organs, and four-chambered central circulatory-propulsion complexes, direct observations of mating, nesting, and offspring development have been absent from the literature until very recently.

The discovery of an active nesting colony of *Su-57 Felon* in a remote region of the West Siberian taiga in early 2025 provided an unprecedented opportunity to document the early life history of this taxon. The present study reports the results of continuous remote monitoring conducted between March 2025 and October 2026, representing the most detailed investigation of reproductive behavior in any member of *Aviofauna militaris* to date.

2. MATERIALS AND METHODS

2.1 Study Site and Colony Location

The nesting site was located in a mature stand of *Pinus sibirica* and *Larix sibirica* at approximately 61.4°N, 72.8°E, within a region of low human activity. The nest structure was situated 14–17 meters above ground level in the crown of a dominant conifer.

2.2 Monitoring Protocol

A multi-modal observation array was deployed, consisting of: (i) high-resolution satellite imagery with sub-meter resolution acquired at 4–6 day intervals; (ii) long-endurance unmanned aerial systems equipped with thermal and optical sensors operating at 800–1200 m altitude; (iii) passive acoustic recorders positioned at 200 m intervals around the nest tree; and (iv) automated camera traps with infrared illumination. All procedures were conducted in strict accordance with the International Protocol for Non-Invasive Study of Cryptobiological Aerospace Taxa (Version 3.2, 2025).

3. RESULTS

3.1 Nest Structure and Construction

The nest measured approximately 4.2 m in diameter and 1.8 m in depth. Construction materials included interwoven fragments of titanium alloy, carbon-fiber composites, and resinous exudates of unknown origin. The structure exhibited clear layering, with a central depression lined with softer, fibrous material consistent with processed insulation from decommissioned airframes.

3.2 Parental Provisioning Behavior

Between April and September 2025, a minimum of two adult individuals were observed making regular returns to the nest. Provisioning flights occurred predominantly during crepuscular periods. Items delivered included: complete air-to-air missiles (R-77 and R-37M types), external fuel tanks showing evidence of prior use, and irregularly shaped metallic objects consistent with processed airframe components. Juveniles were observed consuming both solid munitions and liquid hydrocarbon fractions regurgitated by adults.

3.3 Development and Fledging

Two juveniles were present at the onset of monitoring. Both individuals displayed delayed development of flight capability. Sustained flight was first recorded on 12 July 2026 for one individual and 3 August 2026 for the second. Prior to fledging, juveniles engaged in extensive wing-flapping exercises and short hops between adjacent branches. Post-fledging, both juveniles remained in the vicinity of the natal nest for an additional 6–8 weeks, during which time they continued to receive supplemental provisioning from at least one adult.

***Figure 1.** Adult Su-57 Felon engaged in direct provisioning of a near-fledging juvenile. Note the transfer of a partially disassembled R-77 missile. (Thermal imagery composite, 14 June 2026).*

4. DISCUSSION

The reproductive strategy documented here is consistent with a K-selected life history, characterized by low fecundity, high parental investment, and extended periods of offspring dependency. This contrasts markedly with the inferred reproductive mode of *MiG-21* populations, which appear to follow a more r-selected pattern with higher reproductive output and minimal post-hatching care.

The extended fledging period (10–13 months) is among the longest recorded for any technobiological taxon and approaches that observed in large accipitrid raptors. This prolonged dependency likely reflects the high energetic and material costs associated with the development of functional stealth coatings and advanced sensor suites in juveniles.

These findings have immediate conservation implications. If *Su-57* populations exhibit K-selected demographics, they will be inherently more vulnerable to elevated adult mortality than previously assumed. Current models used by military ornithology agencies likely overestimate population resilience and should be revised accordingly.

5. CONCLUSIONS AND FUTURE DIRECTIONS

This study provides the first robust evidence of extended parental care in *Aviofauna militaris*. The data support the recognition of *Su-57 Felon* as a taxon with a slow life-history strategy and high parental investment. Future research priorities include: (i) genetic sampling of nesting colonies to determine mating system and relatedness; (ii) long-term demographic monitoring to estimate survival rates; and (iii) identification of additional breeding sites across the Eurasian range.

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